

The Impartial Realizer Problem: One Mover Bit from Symmetry

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Abstract

The weighted-source WITT-FIFO theorem of the companion flagship paper realizes $\{Q = 0\}$ as a normal-play P -set for every quadratic refinement Q of every alternating form on $\mathbb{F}_2^n$, under a bounded $(w_0, c) = (1, 1)$ access contract. The compiled arena is partizan through exactly one bit: the terminal claim move is keyed to the accumulated charge *and the mover phase*, so which seat may claim depends on whose turn it is; every other move is seat-symmetric. This note isolates the flagship's first residual open question: does an *impartial* rule – identical move sets for both players, ordinary normal play – meet the same access contract and realize the same P -sets uniformly? We collect the proved fence (the coin-turning linear ceiling, the abelian obstruction, the transcript-span bound), record the trivial rigid-tempo compilation lemma, and reduce the design problem to a pass-free or pass-parity-pinned analogue of the matching forcing theorem, which is the open technical content.

1 The problem

Throughout, $V = \mathbb{F}_2^n$, B is an alternating bilinear form on V , and Q is a quadratic refinement of B , i.e. $Q(u+v) = Q(u)+Q(v)+B(u, v)$ with $Q(0) = 0$. The access contract F1–F3/N1–N2, the loading map, and the realization theorem are those of the flagship paper [1]: statics are refinement-blind (F1), each move predicate and transition observes Q through at most c oracle queries at points of Hamming weight at most w_0 , with (w_0, c) independent of dimension (F2), the semantics is one declared outcome convention (F3), and exactness holds uniformly over the full torsor of refinements of every form in the family (N1) within the stated budget (N2).

The WITT-FIFO arena satisfies the contract at $(w_0, c) = (1, 1)$ and has root outcome P iff $Q(x) = 0$ for every loaded input x [1, weighted-source theorem]. Its state carries a coin set, a FIFO queue of open coins, a Boolean ko bit, the accumulated charge σ , and the mover phase ϕ ; the moves are OPEN, CLOSE, PASS, and one terminal claim. OPEN, CLOSE, and PASS are available to whichever player moves. Only the claim is partizan: at a terminal board of charge σ , the unique claim exists exactly when the player to move is the seat designated by σ . The arena is therefore partizan through the single bit ϕ , consumed at one move.

Question 1 (impartial uniform realizer). **[OPEN]** Does there exist an impartial rule – every legal move available to both players, outcomes read by ordinary normal play (last player to move wins) – satisfying F1–F3/N1–N2 at some bounded (w_0, c) , whose root outcome at loaded input x is P exactly when $Q(x) = 0$, uniformly over every quadratic refinement Q of every alternating form B in every finite dimension?

The question is aesthetically loaded for the motivating instance: the Gold trace data $Q_a(x) = \text{Tr}(x x^{2^a})$ is built entirely from *impartial* game arithmetic (nim addition, the Turning-Corners product, its Frobenius diagonal), yet the only known realizer consumes one partizan bit. An affirmative

answer would close the loop – impartial data realized by impartial play; a negative answer would prove the mover bit is the price of the quadric.

2 The proved fence

Three flagship results bound any candidate from below; none decides Question 1.

Theorem 2 (coin-turning linear ceiling). **[PROVED]** *[1, Thm. A] Every Winning Ways coin-turning game has configuration P -set equal to the kernel of an $\mathbb{F}_2$ -linear nimber-valued map; no coin-turning P -set is a quadric of positive polar rank.*

An impartial realizer therefore cannot be a disjunctive sum of coin-turning heaps, and more generally cannot present its positions as configurations whose Grundy value is XOR-additive over independent components. Whatever forces the quadric must entangle the coordinates dynamically, as the FIFO queue does.

Theorem 3 (abelian obstruction). **[PROVED]** *[1, Lem. abelian, Thm. B] In any commutative monoid, squaring is an endomorphism, so any quadratic datum read through homomorphic coordinates has polarization identically zero. In particular no commutative game-value monoid – Sprague–Grundy values under disjunctive sum, misère quotients, turn-parity counters – carries a quadratic refinement with $B \neq 0$.*

This does *not* exclude impartial realizers. It excludes factorizations: an impartial candidate cannot read Q off any commutative value quotient of its positions. The refinement must enter intensionally, through the move relation. Impartial play retains the one asymmetry the flagship identifies as structurally available [1, Rem. screen] – the first-/second-player asymmetry of whose turn it is – even though its move *sets* are equal. Normal play is precisely the reading of that asymmetry.

Theorem 4 (transcript-span bound). **[PROVED]** *[1, transcript-span theorem] Any transcript-stable exact rule observes refinement values at directions spanning the input; in the original frame, $\text{weight} \leq w$ observations number at least $\text{wt}(x)/w$.*

The bound is seat-blind: an impartial realizer at $w_0 = 1$ still needs the full singleton support $\{e_j : x_j = 1\}$. No information discount is available for symmetry.

3 Tempo compilation

The elementary mechanism by which impartial normal play converts a bit into an outcome is play-length parity.

Lemma 5 (rigid tempo). **[STANDARD MATH]** *Let A be a finite impartial arena under normal play in which every maximal play from the root has the same length parity $p \in \mathbb{F}_2$. Then the root outcome is strategy-independent: N if $p = 1$ and P if $p = 0$.*

Proof. The winner of a maximal play of length ℓ is the player who made move ℓ , namely the first player iff ℓ is odd. If every maximal play has $\ell \equiv p$, the winner's identity does not depend on either player's choices. $\square$

Corollary 6 (tempo compilation). **[STANDARD MATH]** *Suppose for each (B, Q, x) an impartial arena with refinement-blind statics and (w_0, c) -metered transitions has the property that every maximal play from the loaded root has length $\equiv Q(x) \pmod{2}$. Then it is an impartial uniform realizer in the sense of Question 1.*

The content is thus a *forced-length* theorem: the total move count parity must equal the charge under both players’ arbitrary play, while transitions stay inside the oracle meter. The flagship arena is close to length-rigid already – every loaded coin is touched exactly twice (one OPEN, one CLOSE), contributing a fixed even base – and its charge σ is public state accumulated during play, so a terminal tail of forced length $\sigma \bmod 2$ is a legal impartial gadget (its legality reads the position, not the oracle). Two mechanisms break rigidity:

1. *PASS*. The flagship strategy uses PASS at opponent checkpoints and as ko relief; each PASS flips play-length parity. The partizan claim is precisely the device that makes pass parity harmless: keying the claim to the mover phase ϕ hands the last move to the correct seat regardless of how many passes occurred.
2. *The claim itself*. As a partizan right it consumes ϕ ; as an impartial move (“claim legal iff $\sigma = 1$, for whoever moves”) it hands the win to whichever player *reaches* the terminal board on their turn, which reintroduces the tempo battle the claim was compiled to avoid.

Remark 7 (the crux). **[OPEN]** The open technical content of Question 1 is therefore a pass-controlled analogue of the flagship’s matching forcing theorem [1, matching FIFO theorem]: a both-seat strategy theorem on matching-plus-isolates boards in which either (i) PASS is eliminated and the safe-front forcing survives with rigid length, or (ii) pass parity is itself strategically pinned, so that total length parity, not claim rights, compiles the charge. The ko rule is the immediate obstruction to (i): in the FIFO transition system a forced pass is what clears ko, and the kernel-checked strategy semantics of the matching theorem uses it. Any replacement must re-prove termination, the queue invariants, and the overlap-forcing lemma under the modified move set – the analogues of what `formal/Ogdoad/FifoMatching.lean` and `formal/Ogdoad/GoldSemantics.lean` check for the partizan compilation.

Remark 8 (weaker targets). Two retreats would still be informative. A realizer at larger but bounded (w_0, c) would separate the impartial question from the weight-one optimality question. A negative result – no impartial rule in the declared access class realizes the torsor uniformly – would prove the mover bit necessary and give the flagship’s access contract a genuinely partizan floor; by Theorem 3 such a proof cannot proceed through value quotients and must engage the move relation directly, plausibly through a parity-transport invariant of impartial transition systems.

4 Status

Everything in Section 2 is proved in the flagship paper; the lemmas of Section 3 are trivial and included to fix the compilation target. Question 1 is open in both directions. This note is the authoritative statement of the problem and will be folded into the flagship paper when resolved.

References

- [1] a9lim, *Quadratic refinements in normal play: realization and the observation boundary*, companion flagship paper, `writeups/goldarf.tex` in the ogdoad repository, 2026.

- [2] E. R. Berlekamp, J. H. Conway, R. K. Guy, *Winning Ways for Your Mathematical Plays*, 2nd ed., A K Peters, 2001–2004.
- [3] T. S. Ferguson, *Game Theory*, Part I, Ch. 5: Coin Turning Games, UCLA online lecture notes.
- [4] T. E. Plambeck, A. N. Siegel, *Misere quotients for impartial games*, J. Combin. Theory Ser. A **115** (2008), 593–622.